

IPV4 & IPV6

Module 12

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Video: <https://www.youtube.com/watch?v=HXbCUPDXLG8>

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana I welcome you all to the lecture series of cyber and information security in this lecture we will study about ipv4 and IPv6 protocols used at Network layer level first of all please see the contents which I shall cover after covering brief introduction about Internet Protocol version 4 I shall take up header of ipv4 which will be followed by addressing in ipv4 in last I shall take up internet protocol version 6 that is ipv 6 so now let us start Internet Protocol version 4 that is ipv4 is the fourth version in the development of the Internet Protocol it is one of the core Protocols of Internet

it still Roots most internet traffic today despite the ongoing development of the successor protocol IPv6 ipv4 is described in ietf publication RFC 791 in September 1981 replacing an earlier definition RFC 760 in January 1980 ipv4 is a connectionless protocol for use on packet switch networks with 30 2bit address now first of all let us see header of ipv 4 and ipv4 datagram consist of a header part and a text part the header has a 20 by fixed part and a variable length optional part the header format is shown in figure there are minimum five rows of 32 bit each additional rows are optional now we shall see fields of ipv4 header one by one version the version field are first four bits in first row of Ip header for IP V4 this has a value of four other versions

are ipv5 which was an experimental realtime stream protocol that was never widely used and IPv6 which is expected to be deployed in near future internet header length IHL this field of 4 bits is the internet header length IHL which indicates number of 32bit rows in the header the minimum value is five which applies when no options are present the maximum value of this 4bit field is 15 which limits the header to 60 bytes and thus the option field to 40 bytes differentiated Services code point dscp this field of six bits is now defined for differentiated Services new technologies are emerging that require realtime data streaming and therefore make use of the dscp field an example is voice over IP which

is used for interactive data voice exchange originally this six-bit field was defined as type of service which contained a 3-bit precedence field and three flags delay d throughput t and reliability R explicit congestion notification ecn this field of two bits is now defined which allows end to end notification of network congestion without dropping packets earlier these two bits were not defined total length the total length of 16 bits includes length of both header and data the maximum length is 65,535 bytes identification this field is an identification field and is primarily used for uniquely identifying the group of fragments of a single IP datagram the identification field is needed to allow the destination host to

determine which datagram a newly arrived fragment belongs to all the fragments of a datagram contain the same identification value flag there are three bits in this field out of which one is unused bit and two are one bit Fields DF and MF DF stands for don't fragment and MF for more fragment DF is an order to the routers not to fragment the datagram because the destination is incapable of putting the pieces back together again MF stands for more fragments all fragments except the last one have this bit set it is needed to know when all fragments of a datagram have arrived fragment offset this field of 13bit tells where in the current datagram this fragment belongs all fragments except the last one in a datagram must be a multiple of 8 bytes the elementary

fragment unit since 13 bits are provided there is a maximum of 8192 fragments per datagram giving a maximum datagram length of 65,536 bytes one more than the total length field time to live this field of 8 Bits is a counter used to limit packet lifetime it is supposed to count time in seconds allowing a maximum lifetime of 255 seconds protocol this field of 8 Bits Define the protocol used in the data portion of the IP datagram header check sum this field of 16bit verifies the header only such a check sum is useful for detecting errors generated by bad memory words inside a router Source address this field of 32bits is the IP V4 address of the sender of the packet this address may change in transit by a network address translation device destination address this field of

32bit is the ipv4 address of the receiver of the packet as with the source address this may change in the transit by a network address translation device next field is options field it was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design to permit experimenters to try out new ideas and to avoid allocating header bits to information that is rarely needed originally five options were defined as listed in figure but since then some new ones have been added as shown in figure the security option tells how secret the datagram is the strict Source routing option gives the complete path from source to destination as a sequence of IP addresses the loose Source routing option

gives a list of routers RS not to be missed on the way the record route option tells the router along the path to append their IP address to the option Fields finally the time stamp option is like the record route option except that in addition to recording its 32bit IP address each router

also records a 32bit time stamp second part of Ip header is data portion the data portion of the packet is not included in the packet check sum its contents are interpreted based on the value of the protocol header field some of the common protocols for the data portion are listed in the table here protocol value one indicates icmp two indicates I igmp six indicates TCP 17 indicates UDP 41 indicates en NCAP 89 indicates OSF and 132 indicates

sctp so this was ipv4 header now let us see ipv4 addressing schemes Network addresses in IP P V4 which are 32bit numbers are usually written in dotted decimal notation in this format 32bit IP address is divided into four parts of one byte each each of the four bytes is written in decimal notation from 0 to 255 for example the 32bit 1 0 1 0 1 1 0 0 0 0 0 1 0 0 0 0 1 1 1 1 1 1 1 0 0 0 0 001 address is written as 172.16.254.1 the low lowest IP in this addressing scheme is 0.0.0.0 and the highest IP is 255.255.255.255 for several decades IP addresses were divided into five categories as shown in figure this allocation is class full addressing it is no longer used but references to it in the literature are still common we will discuss the

replacement of class full addressing shortly first let us see classful addressing in classful addressing there are five types of classes the class A starts with zero bit Class B starts with 1 Class C starts with 1 1 0 class D starts with 1 1 1 and Class E starts with 1 1 one 1 there are eight bits for Network address and 24 bits for host address in class A and total address ranges from 1.0.0.0 to 127.255.255.255 in class A There are 16 bits for Network addresses and 16 bits for host addresses in class B and address ranges from 128.0.0.0 to 191.

255.255.255.255 in class B there are 24 bits for networks and eight bits for host address in class C and total address ranges from 192.0.0.0 to 223.255.255.255 in class C Class D addresses which starts with 1110 are used for multicasting Class E addresses which starts with one one one one are reserved for future use now let us see what is Network address and subnet mask net Network address is first IP address of any network or subnet here IP address is 192.128.64.100/24 where /24 indicates that there are 24 bits in network address and rest eight bits are for host so subnet mask contains 24 1s and 8 0s with subnet mask as 255.255.255.0 Network address is obtained by performing and operation of IP address of host with subnet mask as

shown in figure finally Network address obtained is 192.128.64.0 classful addressing prescribed only three possible subnet mask for class A it is 255.0.0.0 or /8 for class B it is 255.255.0.0 or /16 and for Class C it is 255.255.255.0 or /24 this was class full addressing now let us see class L inter domain routing cidr the basic idea behind cidr which is described in RFC 1519 is to allocate the remaining IP addresses in a variable sized blocks without regard to the classes if a site needs say 2,000 addresses it is given a block of 2048 addresses on a 2048 byte boundary to make the forwarding algorithm easier to understand let us consider an example in which millions of addresses are available starting at 194.

24.0 suppose that Cambridge University needs 2048 addresses and is assigned the addresses 194.24.0 through 194.24.7 255 along with subnet mask 255.255.0.0 next Oxford University asked for 4096 addresses since a block of 4096 addresses must lie on a 4096 byte boundary they cannot be given addresses starting at 194.24.8.0 instead they get 194.24.16.0 through 194.24.31 255 along with subnet mask 255.255.0.0 now the University of Edinburgh asked for 1024 addresses and his 19 minutes, 1 second assigned addresses 194.24.80 through 194.24.111 255 along with subnet mask 255.255.252.0 these assignments are summarized in figure after CIDR let us see net which is Network address translation the basic idea behind net is to assign each company a single IP address for internet traffic within the company every computer gets a unique IP address which is used for routing internal traffic however when a packet exit the company and goes to the ISP and address translation takes place to make this scheme possible three ranges of IP addresses have been declared as private companies may use them internally as they wish the Only Rule is that no packet containing these addresses may appear on the internet itself these private addresses are also not used for any websites the following are the three

ranges reserved for private networks as defined in RFC 1918 for class A it is 10.0.0.0 to 10.255.255.255 with 16,777,216 host in each Network for class B it is 172.16.0.0 to 172.31.255.255 with 65,536 host in each Network for Class C this range is 192.168.0.0 to 192.168.255.255 with 2,097,152 host in each Network the operation of net is shown in figure within the company premises each machine has a unique address of the form 10.x.y.z however when a packet leaves the company premises it passes through a net box that converts the internal IP Source address 10.0.0.1 in the figure to the company's true IP global address 198.51.100.12 in given example the netbox is often combined in

a single device with a firewall which provides security by careful control so this was ipv4 now let us move on to ipv6 Internet Protocol version 6 IPv6 was developed by the internet engineering task force ietf to deal with the long anticipated problem of ipv4 address exhaustion IPv6 is intended to replace ipv4 IPv6 addresses are 128 bits which are represented as eight groups of four hexadecimal digits with the groups being separated by colons as shown in figure the main feature of IPv6 are discussed below first and foremost IPv6 has longer addresses than ipv4 which is 128 as compared to 32 the second major Improvement of IPv6 is the simplification of the header it contains only Seven Fields as compared to 13 in IPV4

the third major Improvement was better support for options the fourth area in which IPv6 represents a big Advance is in security finally more attention has been paid to quality of service the IPv6 header is shown in figure an IPv6 p has two parts a header and payload the fixed header occupies the first 40 bytes of the IPv6 packet the version field of 4 bits is always six for IPv6 as it was 4 in case of ipv4 the traffic class field of eight bits is used to distinguish

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between packets with different real time delivery requirements the flow label field of 20 bits is used to allow a source and destination to set up a pseudo connection with particular properties and requirements with regard to flow

control the payload length of 16 bits field tells how many bytes follow the 40 byte header the name was changed CH from ipv4 total length field because the meaning was also changed slightly the 40 header bytes are no longer counted as part of the length this next header field of 8 Bits tells the receiver how to interpret the data which follows the header this field tells which of the six extension header if any follow this one if this header is the last IP header the next header field tells which transport protocol handles to pass the packet to the Hop limit field of 8 Bits is used to keep packets from living forever it is in practice the same as the time to live field in ipv4 namely if field that is decremented on each hop in theory in ipv4 it was time in

seconds but no router used it that way so the name was changed to reflect the way it is actually used next Comes The Source address and destination address fields which are of 16 bytes each and contain source and destination address a new notation has been devised for writing 16 byte addresses they are written as eight groups of four hexadecimal digits with columns between the groups as shown in Figure 8 0000 0 colon 0000 0 colon 0 0000 0 colon 000000 0 colon 0123 colon 4567 colon 89 a b column CDE e f since many addresses will have many zeros inside them optimizations have been authorized first leading zeros within a group can be omitted say 0 1 2 3 can be written as 1 2 3 second one or more group of 160 bits

can be replaced by a pair of columns thus the above addresses now become 8 00 colon colon 1 2 3 colon 4567 colon 89 a B colon CDE e f now let us see extension headers in IPv6 some of the missing ipv4 fields are occasionally still needed so ipv4 has introduced the concept of an optional extension header these headers can be supplied to provide extra information but encoded in an efficient way six kinds of extension headers are defined at present as listed in figure The Hop by hop header is used for Miscellaneous information for routers destination options provides additional information for the destination a router field provides list of routers to visit fragmentation is for management of datagram

fragments authentication deals with verification of the sender's identity encrypted security payload provides information about the encrypted contents now after options let us see comparison of fields in ipv4 and IPv6 header the IHL field is gone in IPv6 because the IPv6 header has a fixed length header the protocol field was taken out in IPv6 because the next header field tells what follows the last IP header all the fields related to fragmentation were removed because IPv6 takes a different approach to fragmentation for time to live in ipv4 there is a hop limit in ipv6 finally the checksum field is gone because calculating it greatly reduces performance so friends this was ipv4 and ipv6 finally summarizing we learned about introduction to Internet Protocol

which was followed by header of ipv4 and details of addressing in ipv4 in the end we discussed IPv6 header hope the concepts explained in this lecture were understandable and helpful hope to see you again in next lecture till then goodbye enjoy the day thank you